

From Barnyard to Food Table: the Omnipresence of Hepatitis E virus and Risk for Zoonotic Infection and Food Safety

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Hepatitis E virus (HEV) is an important but extremely understudied pathogen. The mechanisms of HEV replication and pathogenesis are poorly understood, and a vaccine against HEV is not yet available. HEV is classified in the family Hepeviridae consisting of at least four recognized major genotypes. Genotypes 1 and 2 HEV are restricted to humans and associated with epidemics in developing countries, whereas genotypes 3 and 4 HEV are zoonotic and responsible for sporadic cases worldwide. The identification and characterization of a number of animal strains of HEV from pigs, chickens, rabbits, rats, mongoose, deer, and possibly cattle and sheep have significantly broadened the host range and diversity of HEV. The demonstrated ability of cross-species infection by some animal strains of HEV raises public health concerns for zoonotic HEV infection. Pigs are a recognized reservoir for HEV, and pig handlers are at increased risk of zoonotic HEV infection. Sporadic cases of hepatitis E have been definitively linked to the consumption of raw or undercooked animal meats such as pig livers, sausages, and deer meats. In addition, since large amounts of viruses excreted in feces, animal manure land application and runoffs can contaminate irrigation and drinking water with concomitant contamination of produce or shellfish. HEV RNA of swine origin has been detected in swine manure, sewage water and oysters, and consumption of contaminated shellfish has also been implicated in sporadic cases of hepatitis E. Therefore, the animal strains of HEV pose not only a zoonotic risk but also food and environmental safety concerns.